

CERTIFICATE

This to certify that the Mini Project report on **DataSentinel** has been submitted by Chinmay Ghosh (23107001) Subodh Wani (23107070) Shantaram Gawas (23107108) Nishigandha Sawant (23107047) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science & Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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TABLE OF CONTENTS

Abstract	
1. Introduction.....	1
1.1.Purpose.....	2
1.2.Problem Statement.....	3
1.3.Objectives.....	5
1.4.Scope.....	6
2. Literature Review.....	7
3. Proposed System.....	9
3.1 Features and Functionality.....	10
4. Requirements Analysis.....	12
5. Project Design.....	15
5.1.Use Case diagram.....	16
5.2.DFD (Data Flow Diagram)	17
5.3.System Architecture.....	19
5.4.Implementation.....	21
6. Technical Specification.....	26
7. Project Scheduling.....	31
8. Results.....	35
9. Conclusion.....	37
10. Future Scope.....	38
References	

ABSTRACT

DataSentinel is an AI-powered Data Quality Intelligence System designed to democratize advanced data analysis through a no-code, integrated platform. In the era of exponential data growth, raw datasets often contain inconsistencies, missing values, and anomalies that significantly degrade the reliability of machine learning models and analytics pipelines.

The proposed system integrates statistical profiling, machine learning-based anomaly detection, and natural language interaction into a unified platform. It employs an ensemble of algorithms—including Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and Z-score—to detect anomalies with enhanced robustness.

A key contribution of DataSentinel lies in its Explainable AI (XAI) framework, which converts complex mathematical outputs into human-readable explanations. Additionally, the system incorporates a natural language chatbot powered by Mistral-7B, enabling users to query datasets conversationally without requiring programming expertise.

By combining data preprocessing, anomaly detection, visualization, and NLP capabilities, DataSentinel significantly reduces the technical barrier in data science workflows, aligning with SDG 9 (Industry & Innovation) and SDG 4 (Quality Education).

Chapter 1

Introduction

The rapid advancement of digital technologies and the widespread adoption of data-driven systems have led to an unprecedented surge in the volume, velocity, and variety of data being generated across industries. From business transactions and social media interactions to scientific research and IoT devices, modern systems continuously produce massive datasets that serve as the foundation for decision-making, predictive analytics, and artificial intelligence applications. While this exponential data growth has enabled organizations to derive valuable insights, it has also introduced significant challenges related to data quality, consistency, and reliability.

In real-world scenarios, datasets are rarely clean or structured. They often contain missing values, duplicate records, inconsistent formats, noise, and outliers that can severely impact the performance of analytical models. The presence of such anomalies leads to biased predictions, inaccurate statistical interpretations, and unreliable machine learning outcomes. Despite the critical importance of data quality, many organizations and individuals still rely on manual data cleaning techniques or fragmented tools that require significant technical expertise and time investment. This creates a major bottleneck in the data science pipeline, especially for users who lack programming knowledge.

Traditional data preprocessing approaches are largely dependent on scripting languages such as Python or R, requiring users to write complex code for tasks like exploratory data analysis (EDA), feature engineering, and anomaly detection. These methods not only demand a steep learning curve but also lack scalability when dealing with large datasets. Furthermore, existing tools often focus on isolated functionalities—either providing statistical analysis without intelligent cleaning or offering machine learning models without interpretability. As a result, there exists a significant gap between raw data acquisition and meaningful, trustworthy insights.

Another critical challenge in modern data systems is the lack of explainability in automated analytical processes.

While machine learning algorithms can effectively detect anomalies and patterns, they often operate as black-box models, providing results without clear justification. This lack of transparency reduces user trust and limits the adoption of such systems, particularly among non-technical users who require intuitive and understandable explanations. Consequently, there is a growing need for systems that not only automate data analysis but also provide human-readable interpretations of their outputs.

This project, DataSentinel: Data Quality Intelligence System, is motivated by the need to bridge this gap by developing a unified, intelligent, and accessible platform for data analysis and preprocessing. The system aims to democratize data intelligence by integrating statistical methods, machine learning algorithms, and natural language processing into a single no-code environment. By enabling users to upload datasets, analyze their structure, detect anomalies, and receive actionable insights without writing code, DataSentinel significantly reduces the technical barrier associated with data science workflows.

The proposed system aligns with global initiatives such as Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by promoting scalable and intelligent data processing systems, and SDG 4 (Quality Education) by serving as an interactive learning platform for students and aspiring data analysts. Through its intuitive interface and explainable outputs, DataSentinel empowers users to understand their data more effectively and make informed decisions based on reliable insights.

1.1 Purpose:

The primary purpose of DataSentinel is to develop a centralized and intelligent platform that simplifies the process of data analysis, cleaning, and anomaly detection for users with varying levels of technical expertise. The system is designed to provide an end-to-end solution that transforms raw, unstructured datasets into high-quality, analysis-ready data through automated processes and intelligent recommendations.

Firstly, the platform aims to provide a comprehensive data profiling mechanism that enables users to understand the structure and characteristics of their datasets. This includes identifying missing values, detecting duplicates, analyzing distributions, and recognizing patterns within the data.

Secondly, DataSentinel incorporates advanced machine learning techniques for anomaly detection using an ensemble approach. Algorithms such as Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and statistical Z-score methods are combined to improve detection accuracy and robustness. This multi-algorithm strategy ensures that both global and local anomalies are identified, addressing the limitations of single-model approaches.

In addition to analytical capabilities, the system integrates a natural language processing (NLP) interface powered by large language models. This allows users to interact with their datasets using simple conversational queries, eliminating the need for coding knowledge. Users can ask questions such as “What are the missing values?” or “Which columns contain outliers?” and receive context-aware responses.

Another key purpose of the system is to provide Explainable AI (XAI) outputs, where complex mathematical results and anomaly scores are translated into human-readable explanations. This enhances transparency, builds user trust, and facilitates better decision-making.

By achieving these objectives, DataSentinel aims to create an accessible, scalable, and intelligent platform that bridges the gap between raw data and meaningful insights.

1.2 Problem Statement:

The increasing complexity of datasets and the growing reliance on automated analysis tools have made it difficult for users to ensure data reliability before deriving insights. Without proper validation and understanding of data quality, even advanced analytical processes can produce misleading outcomes. This highlights the need for an integrated system that simplifies data understanding while maintaining accuracy and transparency.

Target Audience:

The primary users of DataSentinel include data analysts, students, researchers, and business professionals who work with datasets but may not possess advanced programming skills. This audience often relies on traditional tools such as spreadsheets or manual scripts, which are inefficient for handling large and complex datasets. Additionally, many users struggle to interpret statistical outputs and machine learning results due to a lack of domain expertise.

The Critical Gap:

This further emphasizes the need for a system that not only simplifies data analysis but also ensures clarity, reliability, and ease of use. By addressing these challenges, users can better interpret their data and make informed decisions without being hindered by technical limitations.

Current data analysis solutions are either highly technical or functionally limited. Advanced tools like Python-based libraries require coding expertise, while simpler tools lack intelligent features such as automated anomaly detection and explainability. Moreover, most existing systems do not provide a unified environment that integrates data profiling, cleaning, anomaly detection, and user interaction, forcing users to rely on multiple disconnected tools and workflows.

Another major limitation is the lack of explainability in machine learning models. While these models can identify anomalies, they fail to provide clear reasoning behind their predictions, making it difficult for users to trust and validate the results. Furthermore, there is minimal support for natural language interaction, forcing users to rely on complex queries or scripts. This creates a significant usability gap, particularly for beginners and non-technical users who require intuitive and interpretable insights.

Our Solution:

DataSentinel addresses these challenges by providing a unified, no-code platform that integrates statistical analysis, machine learning, and natural language processing. Users can upload datasets in various formats, perform automated data profiling, and detect anomalies using an ensemble of algorithms. The system generates detailed insights along with human-readable explanations, ensuring transparency and usability while reducing the need for manual intervention.

Additionally, the inclusion of a conversational interface allows users to interact with the system intuitively, making advanced data analysis accessible to a broader audience.

The platform also ensures a streamlined workflow where all analytical processes are connected within a single environment, improving efficiency and reducing complexity.

1.3 Objectives:

The DataSentinel project is designed to address key challenges in data preprocessing, analysis, and interpretability by developing a unified and intelligent platform. The objectives focus on simplifying complex data operations, improving data quality, and enabling efficient interaction with datasets through modern technologies. The system aims to combine automation, scalability, and explainability to make advanced data analysis accessible to users with varying levels of technical expertise.

The project aims to achieve the following key objectives:

1. Context-Aware Data Preprocessing:

To design and implement a context-aware data preprocessing framework using Python, Pandas, and Scikit-learn that applies domain-informed cleaning strategies to improve data quality, consistency, and reliability of datasets.

2. Query-Driven Data Interaction:

To develop a query-driven data interaction layer using SQL and backend APIs that enables structured dataset modification, transformation, and efficient exploratory data analysis without requiring complex programming.

3. Explainable Anomaly Detection:

To develop an explainable anomaly detection module that produces interpretable outputs, allowing users to understand detected anomalies and validate analytical results with greater transparency and confidence.

4. Scalable Data Analysis Platform:

To build a scalable and reliable data analysis platform capable of processing large datasets efficiently while generating meaningful visualizations to support informed decision-making.

1.4 Scope:

This highlights the importance of a system that ensures clarity and reliability in data analysis, enabling users to make informed decisions more effectively.

The scope of the system includes the following aspects:

1. **Dataset Upload and Processing:**The system supports multiple file formats such as CSV, Excel, and Parquet, enabling users to upload datasets for analysis.
2. **Data Profiling and Quality Analysis:**Automated profiling identifies key characteristics, including missing values, duplicates, and data distributions.
3. **Anomaly Detection:**The platform applies multiple algorithms to detect both global and local anomalies within datasets.
4. **Data Cleaning Recommendations:**Provides intelligent suggestions for handling missing values, removing duplicates, and correcting inconsistencies.
5. **Natural Language Query Interface:**Allows users to interact with the system using simple queries without requiring programming knowledge.
6. **Interactive Visualizations:**Generates charts and graphs to help users understand patterns and relationships in data.
7. **Report Generation:**Enables users to download analysis reports summarizing key findings and recommendations.
8. **Web-Based Accessibility:**The platform is accessible through web browsers, eliminating the need for installation.
9. **Scalability and Future Expansion:**The system is designed to support future enhancements such as real-time data integration, API connectivity, and advanced machine learning models.

Chapter 2

Literature Review

The rapid evolution of data-driven technologies has significantly transformed how organizations and individuals interact with structured and unstructured datasets. With the increasing reliance on data analytics, machine learning, and artificial intelligence, the importance of data quality and preprocessing has become more critical than ever. However, despite advancements in analytical tools, there remains a substantial gap in accessible, intelligent systems that can assist users in understanding and preparing datasets effectively. Recent research highlights that while modern frameworks can process large-scale data efficiently, they often lack usability, interpretability, and integration, particularly for non-technical users.

One of the prominent works in this domain is presented by Li et al. (2025), who introduced the CoTA benchmark along with Adaptive Conversation Reflection (ACR) to evaluate large language models in multi-turn tabular data analysis tasks [1]. However, the study primarily focuses on evaluation metrics within a controlled multi-agent sandbox environment and does not provide a deployable system for real-world data cleaning or preprocessing. As a result, while it advances research in conversational data analysis, it does not address the practical challenges of building an integrated, user-friendly platform for dataset intelligence.

Similarly, Zhu et al. (2026) proposed TiInsight, a SQL-based automated exploratory data analysis (EDA) system that leverages large language models for hierarchical data context generation, text-to-SQL translation, and visualization [2]. The system achieved an impressive 86.3% accuracy on the Spider dataset, demonstrating the potential of LLMs in automating query generation and data exploration tasks. Despite its effectiveness, TiInsight is heavily dependent on SQL-based interactions, which limits its accessibility for users unfamiliar with database query languages. Furthermore, the system lacks the integration of machine learning models for intelligent anomaly detection and data quality assessment, making it insufficient as a comprehensive data preprocessing solution.

In another significant contribution, Agrawal et al. (2025) developed a machine learning framework for automated data cleaning and anomaly detection in large datasets [3]. Their approach integrates Isolation Forest for anomaly detection, Principal Component Analysis (PCA) for dimensionality reduction, and Random Forest algorithms for data imputation. The framework achieved a precision of 0.88 and a Mean Absolute Error (MAE) of 0.015, indicating strong performance in identifying and correcting data inconsistencies. However, despite its technical robustness, the system lacks a natural language interface and does not provide explainability for its predictions. This limits its usability for non-technical users who require intuitive insights rather than raw model outputs.

The existing systems can broadly be categorized into LLM-based analytical systems, SQL-driven EDA platforms, and machine learning-based data cleaning frameworks. While each category addresses specific aspects of data analysis, they remain fragmented and fail to provide a unified solution. LLM-based systems improve interaction but lack deep analytical capabilities, SQL-based systems automate querying but require technical expertise, and ML-based frameworks provide accurate results but lack interpretability and user-friendly interfaces.

Despite these advancements, key research gaps persist in the domain of data quality intelligence. Existing solutions often require programming knowledge, lack explainability, and fail to integrate multiple functionalities within a single platform. Additionally, most systems do not provide a seamless workflow that connects data ingestion, profiling, cleaning, anomaly detection, and insight generation.

To address these limitations, the proposed system, DataSentinel, introduces a unified, no-code data intelligence platform that integrates statistical profiling, ensemble-based anomaly detection, and natural language interaction. Furthermore, Explainable AI enables the conversion of complex outputs into human-readable insights, improving usability and trust. This integrated approach allows DataSentinel to bridge the gap between raw data and actionable insights in a scalable and user-friendly manner [1][2][3].

Chapter 3

Proposed System

The proposed system, DataSentinel: Data Quality Intelligence System, is a web-based platform designed to provide comprehensive data analysis, preprocessing, and anomaly detection through a centralized and user-friendly interface. It addresses the inefficiencies and limitations of existing data analysis tools by integrating statistical methods, machine learning algorithms, and natural language processing capabilities into a unified environment. The system is specifically designed to make advanced data intelligence accessible to non-technical users while maintaining robustness and scalability for large datasets.

DataSentinel offers a unified ecosystem that combines automated data profiling, intelligent data cleaning, multi-algorithm anomaly detection, and interactive visualization. The system processes user-uploaded datasets (CSV, Excel, and other formats) using Pandas-based data pipelines to identify missing values, duplicates, inconsistencies, and structural issues. It further applies an ensemble of machine learning algorithms—including Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and statistical Z-score—to detect anomalies and unusual patterns within the data. Additionally, the platform integrates a natural language interface powered by large language models, enabling users to interact with their datasets through conversational queries. By consolidating these functionalities into a single platform, DataSentinel eliminates the need for multiple fragmented tools and provides users with immediate, actionable insights for data-driven decision-making.

The platform operates on a modular monolith architecture consisting of a Streamlit-based frontend for user interaction, a Python-based backend for processing and orchestration, and dedicated AI/ML modules for data analysis. The storage layer integrates PostgreSQL for metadata management and MinIO for dataset persistence, while Redis is used for caching and performance optimization. This architecture ensures fast response times, scalability, and maintainability, while also supporting explainable outputs that enhance user understanding and trust in the system.

3.1 Features and Functionality:

1. **Dataset Upload and Ingestion:** Users can upload datasets in multiple formats such as CSV, Excel, and Parquet through an intuitive web interface. The system automatically detects file structure, validates schema, and loads data into memory-efficient DataFrames for further processing.
2. **Automated Data Profiling:** The system performs comprehensive exploratory data analysis (EDA), including statistical summaries, data type detection, missing value analysis, duplicate detection, and distribution analysis. It generates insights that help users understand the structure and quality of their datasets.
3. **Data Quality Scoring:** DataSentinel computes a quantitative data quality score based on parameters such as completeness, consistency, uniqueness, and validity. This score provides users with an overall assessment of dataset reliability.
4. **Anomaly Detection (Ensemble Approach):** The platform implements multiple anomaly detection algorithms, including Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and Z-score methods. By combining these techniques, the system ensures robust detection of both global and local anomalies across different data distributions.
5. **Explainable AI (XAI) Outputs:** Unlike traditional black-box systems, DataSentinel provides human-readable explanations for detected anomalies. It translates mathematical scores and model outputs into intuitive descriptions, helping users understand why specific data points are flagged.
6. **Interactive Data Visualization:** The system integrates Plotly and Seaborn to generate interactive visualizations such as histograms, box plots, scatter plots, and correlation matrices. These visual tools enable users to explore patterns, trends, and relationships within their data.
7. **Intelligent Data Cleaning Recommendations:** Based on profiling results, the system suggests appropriate data cleaning strategies, such as handling missing values, removing duplicates, normalizing data, and correcting inconsistencies. Users can apply these transformations with a single click.

8. Natural Language Query Interface: The platform incorporates an NLP-based chatbot powered by Mistral-7B, allowing users to interact with datasets using simple queries such as “Show missing values” or “Identify outliers.” This eliminates the need for coding knowledge.
9. Real-Time Processing and Feedback: The system ensures fast execution of data analysis tasks, providing near real-time feedback for dataset profiling and anomaly detection, enabling efficient decision-making.
10. Report Generation and Export: DataSentinel generates comprehensive reports summarizing dataset insights, anomaly detection results, and cleaning recommendations. Users can download these reports in multiple formats for documentation and further analysis.
11. User-Friendly Interface: The Streamlit-based interface is designed for ease of use, featuring intuitive navigation, minimal setup requirements, and responsive design, making it accessible to users with diverse technical backgrounds.
12. Feedback and Continuous Improvement Mechanism: The system includes a feedback module where users can provide input on analysis accuracy and system performance. This feedback can be used to refine models and improve system functionality over time.

Chapter 4

Requirements Analysis

The requirements analysis defines the functional and non-functional specifications necessary for DataSentinel: Data Quality Intelligence System to effectively serve its target users and achieve its objectives. The system is designed to provide a seamless, intelligent, and scalable platform for data analysis, preprocessing, and anomaly detection, ensuring accessibility for both technical and non-technical users.

4.1 Functional Requirements

1. User Access and Session Management: Users can access the platform directly through web browsers without mandatory account creation, allowing them to perform data analysis tasks quickly and efficiently. The system maintains session-based data handling to ensure temporary persistence of uploaded datasets during analysis. For extended usage, optional user authentication mechanisms can be incorporated to securely manage user sessions and stored reports while maintaining data privacy.

2. Dataset Upload and Ingestion Module: The system allows users to upload datasets in common formats such as CSV, Excel, and Parquet, with appropriate size limits to ensure efficient processing. Upon upload, the system performs automatic file format detection, schema validation, and data loading into structured DataFrames. Additionally, the module provides initial feedback regarding dataset structure, including number of records, attributes, and detected data types, enabling users to verify the dataset before proceeding further.

3. Data Profiling and Analysis Module: The platform performs comprehensive exploratory data analysis (EDA) on uploaded datasets. This includes generating statistical summaries such as mean, median, standard deviation, and quartiles, along with identifying missing values, duplicate entries, and data type inconsistencies. The system presents these insights in a structured and visually interpretable format, enabling users to understand dataset characteristics before further processing.

4. **Data Quality Assessment Module:** The system evaluates dataset quality using predefined metrics such as completeness, consistency, uniqueness, and validity. Based on these parameters, it generates a quantitative data quality score along with detailed insights into potential issues. This module helps users identify critical areas requiring cleaning or preprocessing.

5. **Anomaly Detection Module:** The platform implements an ensemble-based anomaly detection approach using algorithms such as Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and Z-score methods. The system processes numerical and categorical features to identify both global and local anomalies. It provides anomaly scores, classifications, and visual representations to highlight unusual patterns in the dataset.

6. **Explainable AI (XAI) Module:** The system generates human-readable explanations for anomaly detection results and data quality assessments. It translates mathematical computations and model outputs into intuitive descriptions, enabling users to understand the reasoning behind detected anomalies and recommendations without requiring technical expertise.

7. **Data Cleaning and Transformation Module:** The platform provides intelligent recommendations for handling missing values, removing duplicates, normalizing data, and correcting inconsistencies. Users can apply suggested transformations through an interactive interface, with before-and-after comparisons to ensure transparency and validation of changes.

8. **Natural Language Query Interface:** The system integrates a chatbot powered by a large language model (Mistral-7B), allowing users to interact with datasets using simple natural language queries. Users can request insights such as “Show missing values,” “Find anomalies,” or “Summarize dataset,” and receive context-aware responses.

9. **Visualization Module:** The platform generates interactive visualizations using libraries such as Plotly and Seaborn. These include histograms, box plots, scatter plots, correlation matrices, and distribution charts, enabling users to explore patterns and relationships within their data effectively. The visualizations are dynamically generated based on dataset characteristics and support interactive features such as zooming, filtering, and hover-based insights. This enhances user understanding by allowing intuitive exploration of trends, outliers, and correlations, thereby improving decision-making and analytical interpretation.

10. **Report Generation Module:** The system compiles comprehensive analysis reports that include data profiling summaries, anomaly detection results, data quality scores, and cleaning recommendations. Users can download these reports in multiple formats for documentation and further analysis.

11. **User Feedback System:** The platform includes a feedback mechanism where users can provide input on system performance, accuracy of anomaly detection, and usability. This feedback is used to improve model performance and enhance system functionality over time.

4.2 Non-Functional Requirements

Performance:

The platform is designed for high efficiency and responsiveness, ensuring smooth and fast data processing. Dataset profiling and anomaly detection operations are optimized to complete within a few seconds for small to medium-sized datasets. The system ensures minimal latency in generating visualizations and reports, allowing users to receive near real-time insights. Additionally, the overall platform response time for user interactions remains under 2–3 seconds to maintain usability and efficiency.

Reliability:

DataSentinel is designed to provide consistent and dependable performance with a target system uptime of 99% or higher. The system incorporates robust error handling mechanisms, including validation checks for uploaded datasets and informative error messages to guide users in case of incorrect inputs. It ensures secure handling of user data by maintaining temporary storage for uploaded datasets and preventing unauthorized access.

Maintainability:

The system is developed with modular architecture principles to ensure ease of maintenance and scalability. Comprehensive documentation is maintained for all modules, including data processing pipelines and machine learning components. Version control systems and testing frameworks are used to ensure code reliability and facilitate collaborative development. Additionally, the platform supports periodic updates and model retraining to improve system performance.

Chapter 5

Project Design

The design of DataSentinel: Data Quality Intelligence System focuses on creating a unified, intelligent, and user-friendly platform that integrates multiple components to ensure seamless data analysis, preprocessing, and anomaly detection. This chapter presents the structural and functional blueprint of the system, illustrating how different modules—such as dataset ingestion, data profiling, anomaly detection, visualization, and natural language interaction—work cohesively within the platform.

The project design emphasizes scalability, modularity, and usability, ensuring that users can easily upload datasets, analyze their structure, detect anomalies, and derive meaningful insights without requiring programming expertise. The system is designed using a modular monolith architecture, where each component operates independently while maintaining efficient communication with other modules. The design of DataSentinel: Data Quality Intelligence System is centered on establishing a unified, intelligent, and highly user-friendly platform. This system is guided by an architectural philosophy built on three pillars: scalability, modularity, and usability. Scalability ensures performance under increasing data volumes and users through a decoupled architecture. Modularity allows for independent development and maintenance of components, accelerating feature delivery. Usability aims to democratize data quality analysis by providing an intuitive platform for sophisticated analysis without requiring prior programming expertise.

The intelligence core is the Anomaly Detection Module, which implements a suite of advanced machine learning algorithms to automatically identify various anomalies like outliers, temporal drift, and structural errors. The Visualization Module translates these complex profiles and results into intuitive graphical representations, providing interactive dashboards and heatmaps. Finally, the Natural Language Interaction (NLI) Module is a novel component that allows users to query the system and receive insights using plain human language, making data quality analysis highly accessible.

5.1 Use Case Diagram

The use case diagram for DataSentinel illustrates the interactions between the primary actors and the system, capturing the core functionalities of the platform. The main actor is the User, who interacts directly with the system to perform data analysis tasks and obtain insights.

The use case diagram provides a high-level representation of system functionality by illustrating how users interact with different components to perform data analysis tasks.

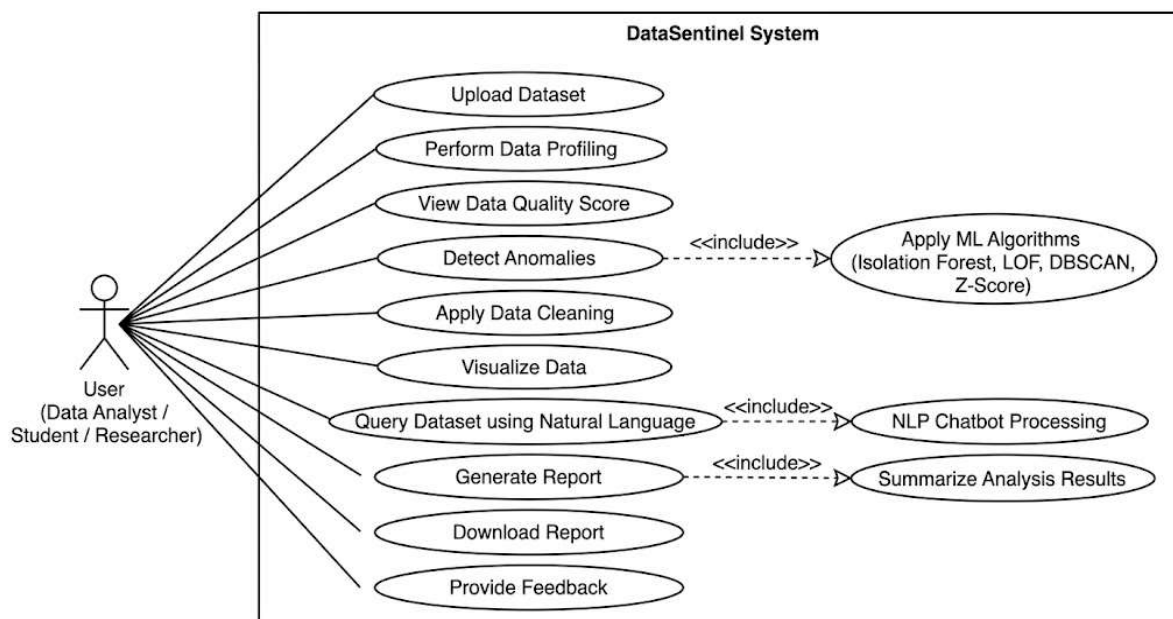


Fig 5.1 DataSentinel use case diagram

Above Fig. 5.1 illustrates the Use Case Diagram of DataSentinel, showing the interaction between the user and various system functionalities. The user can perform several key actions within the system, including uploading datasets in various formats to initiate the analysis process. Once uploaded, the system enables data profiling to understand dataset structure and characteristics, followed by anomaly detection using machine learning algorithms and visualization of patterns through interactive charts.

Additionally, users can interact with the system using natural language queries to retrieve insights without writing code. The platform also supports data cleaning operations and report generation, along with a feedback mechanism to improve system performance and

usability. The system ensures a seamless workflow by integrating all operations within a single interface, allowing smooth transitions from data upload to analysis, visualization, and report generation. This integrated design minimizes complexity and enhances usability for users with varying levels of technical expertise.

5.2 DFD (Data Flow Diagram)

The Data Flow Diagram (DFD) illustrates how data moves through the DataSentinel platform, starting from user input and progressing through various processing stages to generate meaningful outputs. It provides a clear representation of data transformations, processes, and storage components within the system.

The Level 0 Data Flow Diagram presents a high-level overview of the system by showing the interaction between external entities and the core system through primary data flows.

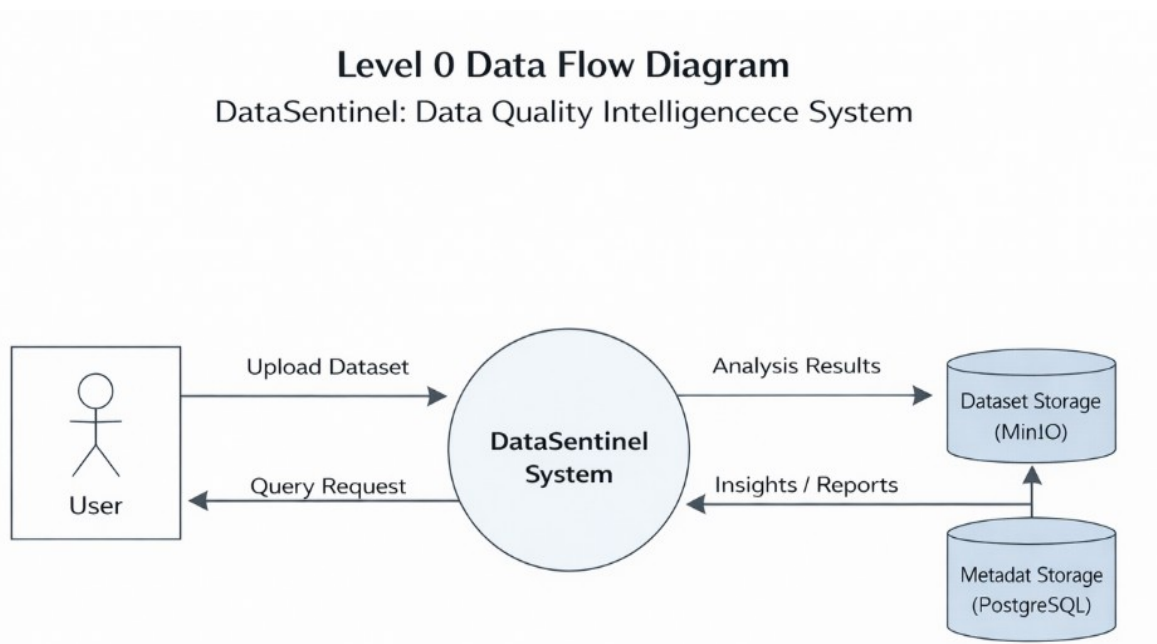


Fig 5.2.1: Data Flow Diagram level 0

Above Fig. 5.2.1 shows the Level 0 Data Flow Diagram, providing a high-level overview of the DataSentinel system and its interaction with external entities. The Level 0 Data Flow Diagram provides a high-level overview of the DataSentinel system, showing the interaction between the user and the system through dataset uploads and query requests.

The system processes this input and generates outputs such as analysis results, insights, and reports, while also interacting with storage components to manage data efficiently.

The Level 1 Data Flow Diagram provides a detailed representation of the internal processes of the DataSentinel system, illustrating how data flows between various modules such as data ingestion, profiling, cleaning, anomaly detection, and visualization, along with interactions with storage components.

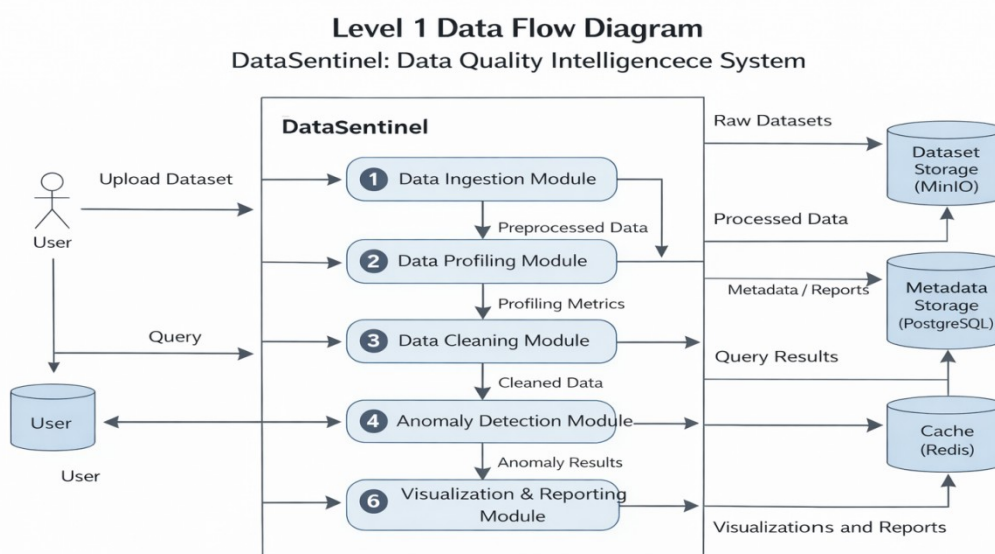


Fig 5.2.2: Data Flow Diagram level 1

Above Fig. 5.2.2 shows the Level 1 Data Flow Diagram, providing a detailed breakdown of internal processes within the DataSentinel system. It expands the main system process into multiple modules, including Data Ingestion, Data Profiling, Data Cleaning, Anomaly Detection, and Visualization & Reporting. The diagram illustrates how data flows sequentially through these modules, starting from dataset upload to final output generation. It also shows interactions with data storage systems such as MinIO for dataset storage, PostgreSQL for metadata management, and Redis for caching. This level of detail helps in understanding how individual components collaborate to perform complex data analysis tasks efficiently.

User Interaction: The process begins when the user uploads a dataset or interacts with the system through the Streamlit interface. This action generates a request that is sent to the backend processing layer.

Data Preprocessing: The uploaded dataset is passed to the data processing engine, where it undergoes validation, preprocessing, and transformation. This includes handling missing values, detecting duplicates, and standardizing formats.

Analysis and Modeling: The processed data is then forwarded to the analysis modules, including the Dataset Analyzer, Data Cleaner, and Anomaly Detection Engine. Machine learning models analyze the data to identify patterns, anomalies, and relationships.

Storage and Retrieval: Intermediate and processed data may be stored temporarily in memory or persisted using storage systems such as PostgreSQL and MinIO. This enables efficient retrieval and reuse of data during analysis.

Output Generation: Finally, the system generates outputs in the form of visualizations, anomaly reports, and data quality scores, which are displayed to the user through the frontend interface.

5.3 System Architecture

The system architecture of DataSentinel defines the structural organization of components and their interactions to enable efficient data processing and analysis. It provides a layered view of how data flows from user input through processing modules to final outputs. This design ensures scalability, modularity, and seamless integration of AI-driven functionalities.

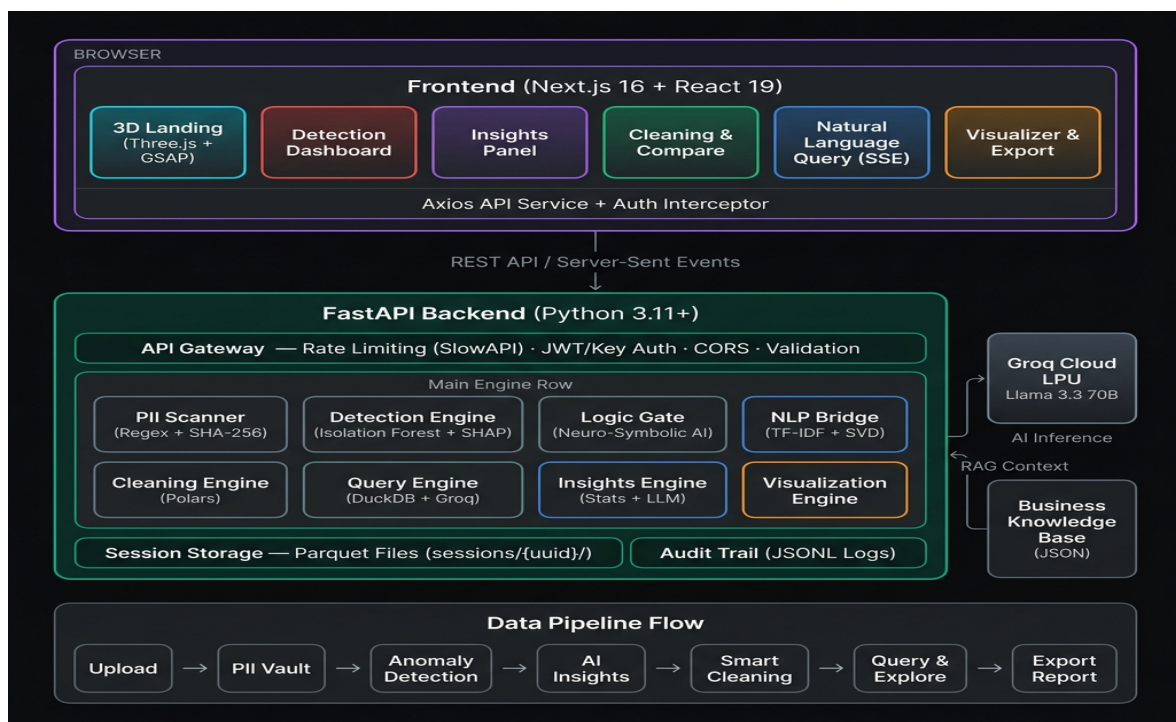


Fig 5.3: System Architecture

Above Fig. 5.3 illustrates the system architecture of DataSentinel, showing how different layers and components interact to perform data ingestion, processing, anomaly detection, and result generation.

5.3.1. Interaction Layer (Frontend)

The Interaction Layer is developed using a **React/Next.js-based frontend**, providing an intuitive and responsive user interface. It allows users to upload datasets, view results, interact with dashboards, and perform natural language queries. The frontend communicates with backend services through REST APIs, ensuring smooth and real-time interaction.

The frontend is designed to be intuitive and responsive, ensuring accessibility across different devices and requiring minimal technical expertise from users.

5.3.2. Processing Layer (Backend Services)

The Processing Layer is implemented using a FastAPI backend, serving as the core computation engine. It includes the anomaly detection engine (Isolation Forest with SHAP), neuro-symbolic logic gate for rule validation, PII scanner for data privacy, natural language query engine using LLMs, and smart data cleaning modules. These components work together to process data efficiently and generate meaningful insights.

1. Dataset Analyzer: Performs statistical profiling and data exploration.
2. Data Cleaner: Applies preprocessing techniques such as handling missing values and removing duplicates.
3. Anomaly Detector: Executes multiple machine learning algorithms to identify anomalies.
4. NLP Chatbot Service: Processes user queries and generates context-aware responses.

This layer ensures efficient handling of requests and maintains communication between the frontend and storage systems.

5.3.3. Storage Layer

The Storage Layer manages data using MinIO for datasets, PostgreSQL for metadata, Redis for caching, and DuckDB for query execution. This ensures efficient data handling, fast retrieval, and scalable performance across the system.

5.4 Implementation

The implementation of DataSentinel follows a modular approach, integrating the frontend, backend, and AI-driven components into a cohesive system. The frontend is developed using a modern React/Next.js framework, while the backend is powered by FastAPI with Python-based data processing and machine learning libraries. The system ensures scalability, efficient performance, and real-time interaction across all modules.

The following system interfaces illustrate key system functionalities, including the landing page, anomaly detection interface, insights dashboard, data cleaning comparison, visualization modules, natural language query system, and final quality reporting. These interfaces collectively demonstrate the end-to-end workflow of DataSentinel, from dataset ingestion and validation to analysis, interpretation, and export of results.

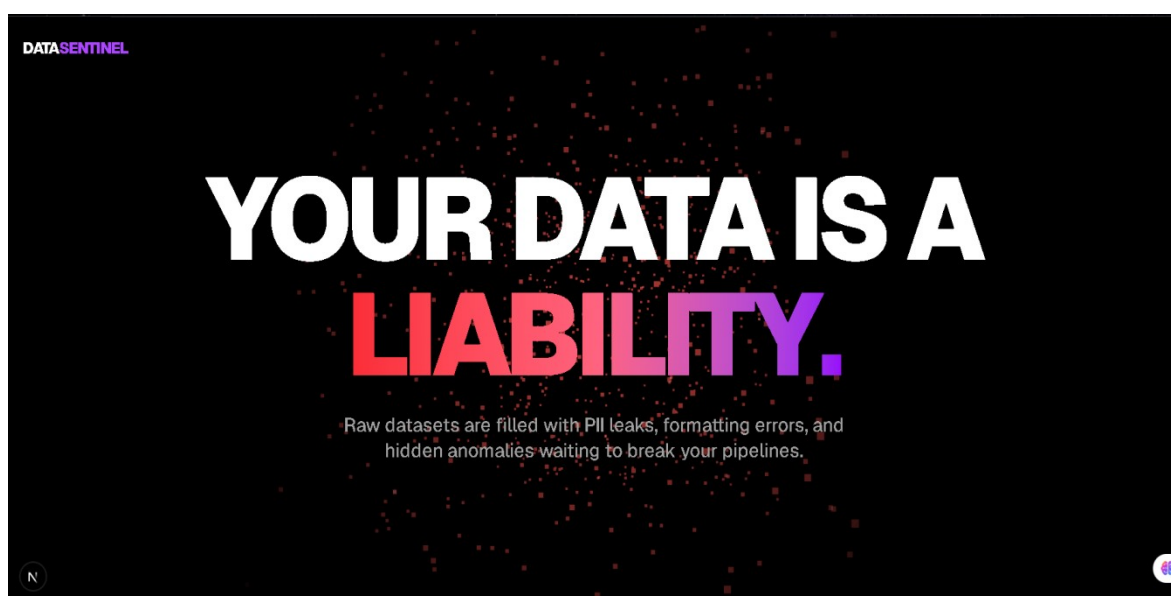


Fig. 5.4.1 Landing page

Above Fig. 5.4.1 shows the DataSentinel landing page, highlighting key data quality issues such as PII leaks and anomalies. The interface is designed to be intuitive and visually engaging for users. It provides a clear overview of system features and offers immediate insights into potential dataset risks.

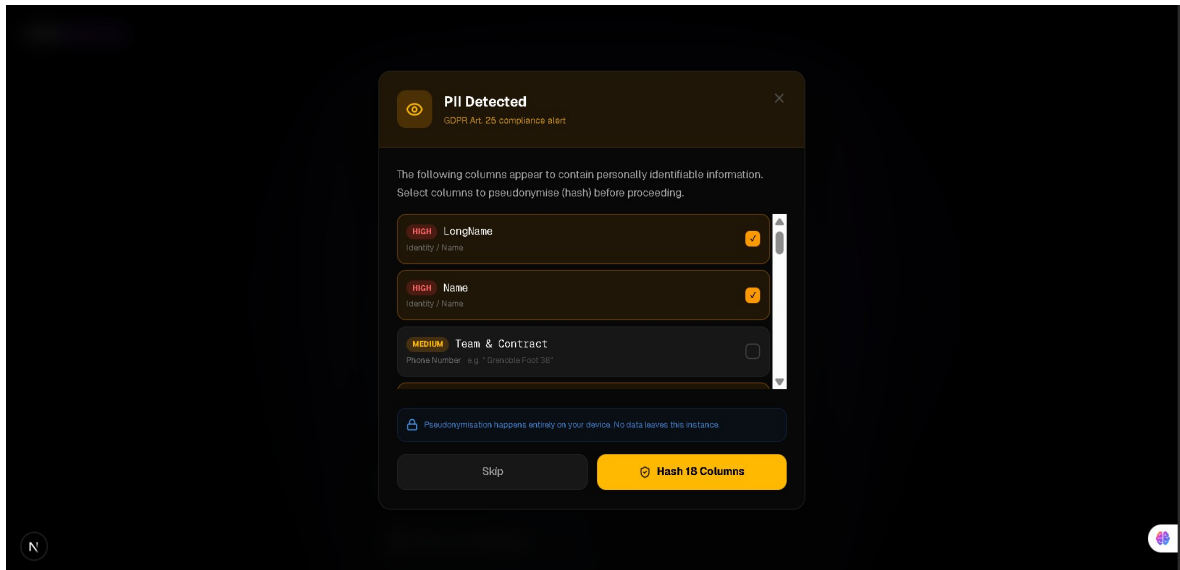


Fig. 5.4.2 Anomaly Detection Engine Results Interface

Above Fig. 5.4.2 shows the anomaly detection results interface with risk scores and SHAP-based explanations. Users can review flagged data, download cleaned datasets, or re-run the pipeline.

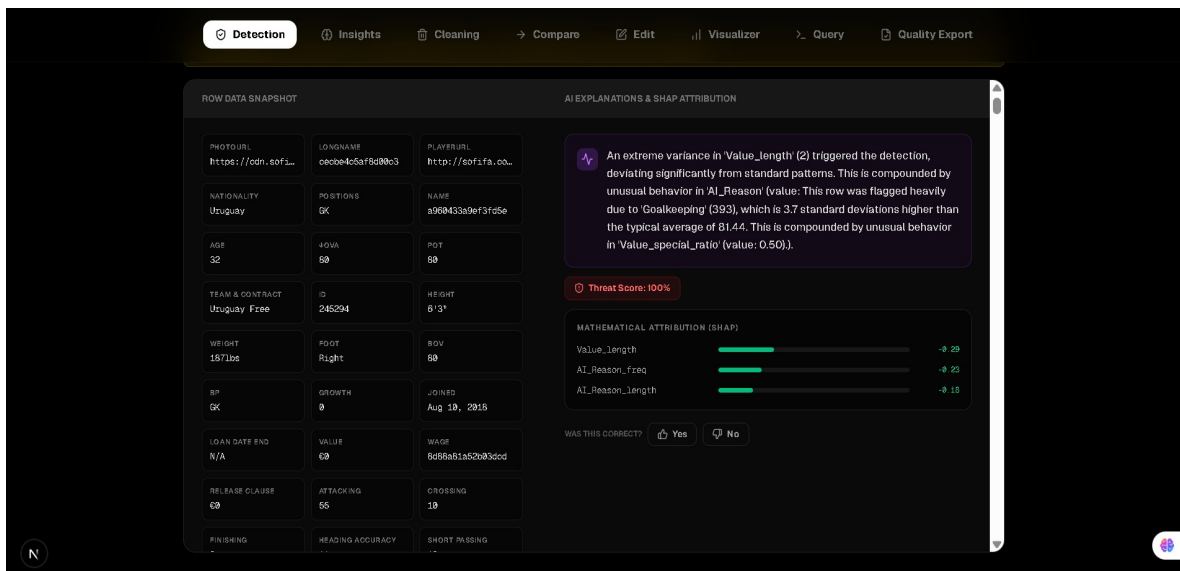


Fig. 5.4.3 Anomaly Detection Results Page

Above Fig. 5.4.3 shows the anomaly detection results page with detected anomalies and associated threat scores. It provides SHAP-based explanations to help users understand the reasoning behind each result. This supports effective validation and interpretation.

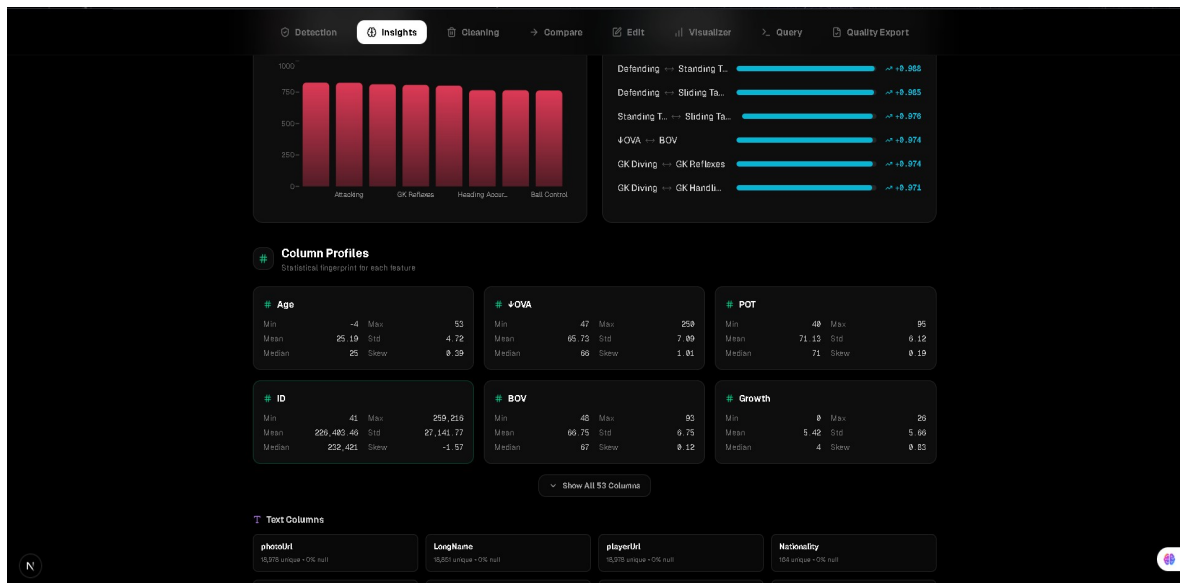


Fig. 5.4.4 Insights Dashboard

Above Fig. 5.4.4 shows the insights dashboard presenting statistical summaries and feature distributions of the dataset. It helps users understand patterns, correlations, and variations within the data. This enables quick interpretation of overall characteristics.

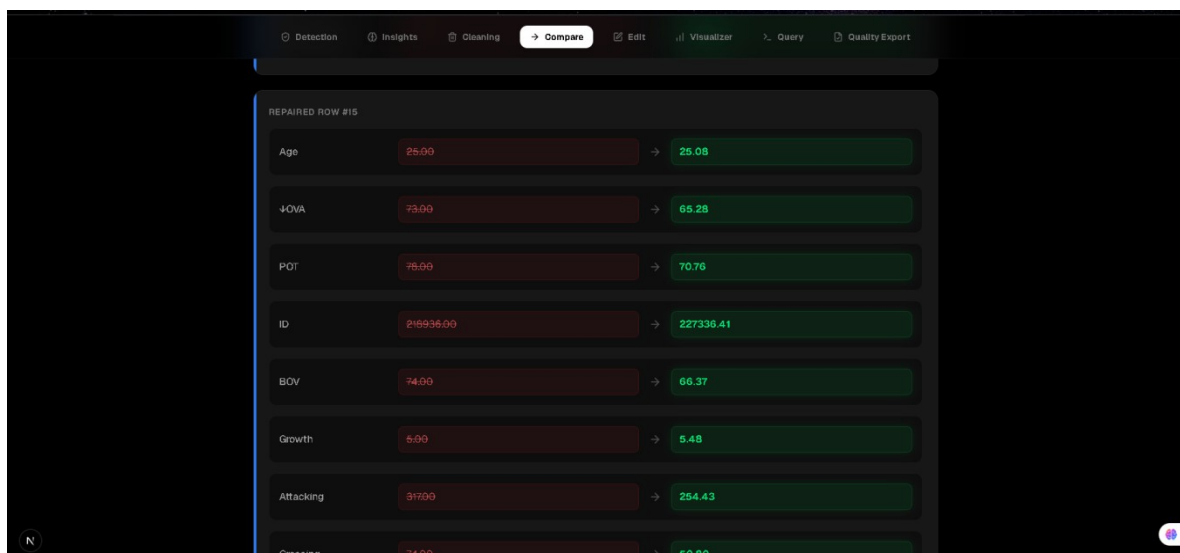


Fig. 5.4.5 Data Comparison Page

Above Fig. 5.4.5 shows the data comparison page highlighting differences between original and cleaned values. Users can track corrections and modifications applied during pre-processing.

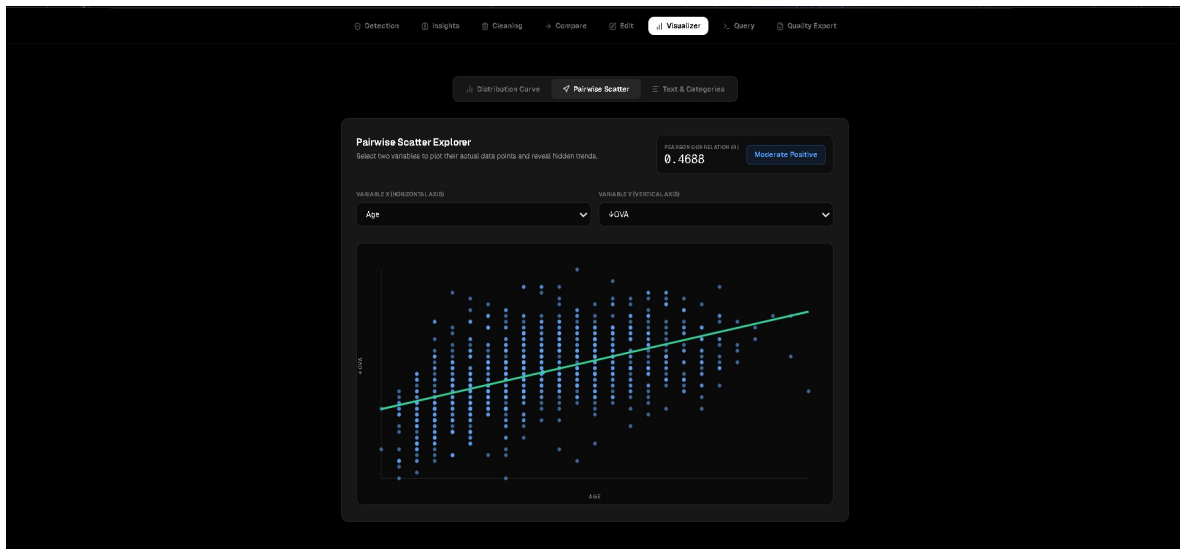


Fig. 5.4.6 Data Visualization Page

Above Fig. 5.4.6 shows the data visualization page with charts such as scatter plots and graphical representations. These visuals help analyze relationships between variables and detect patterns. Users can identify trends and outliers efficiently.

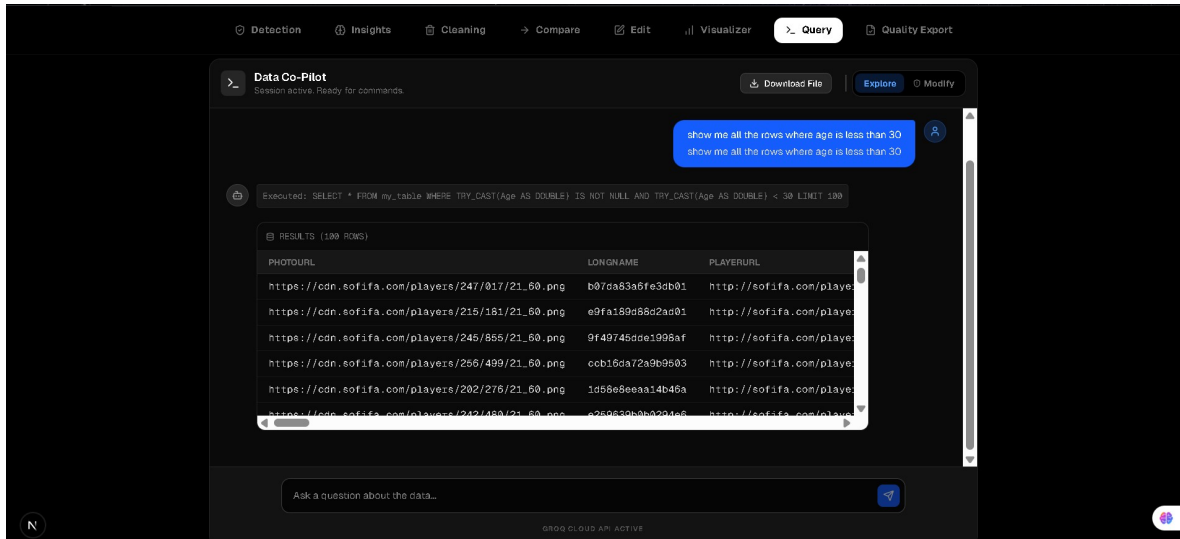


Fig. 5.4.7 Natural Language Query Interface

Above Fig. 5.4.7 shows the natural language query interface for interacting with datasets using simple English queries. The system converts input into SQL and returns results instantly. This simplifies data analysis for non-technical users.

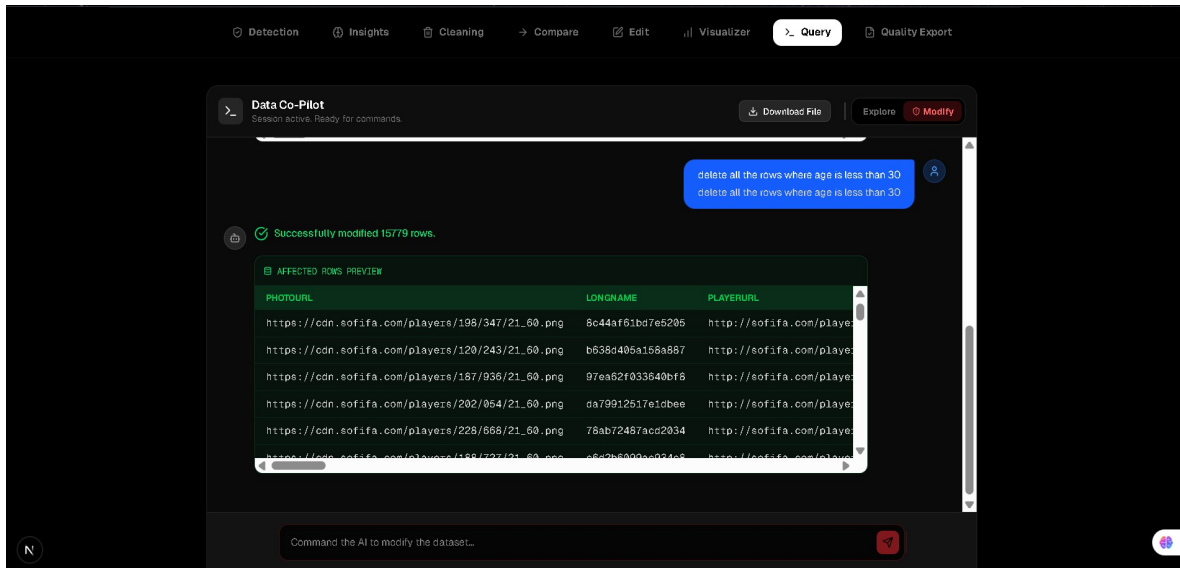


Fig. 5.4.8 Data Modification via Query

Above Fig. 5.4.8 shows real-time data modification based on user-defined queries within the interface. It displays affected rows and confirms successful updates after execution. This supports efficient dataset transformation.

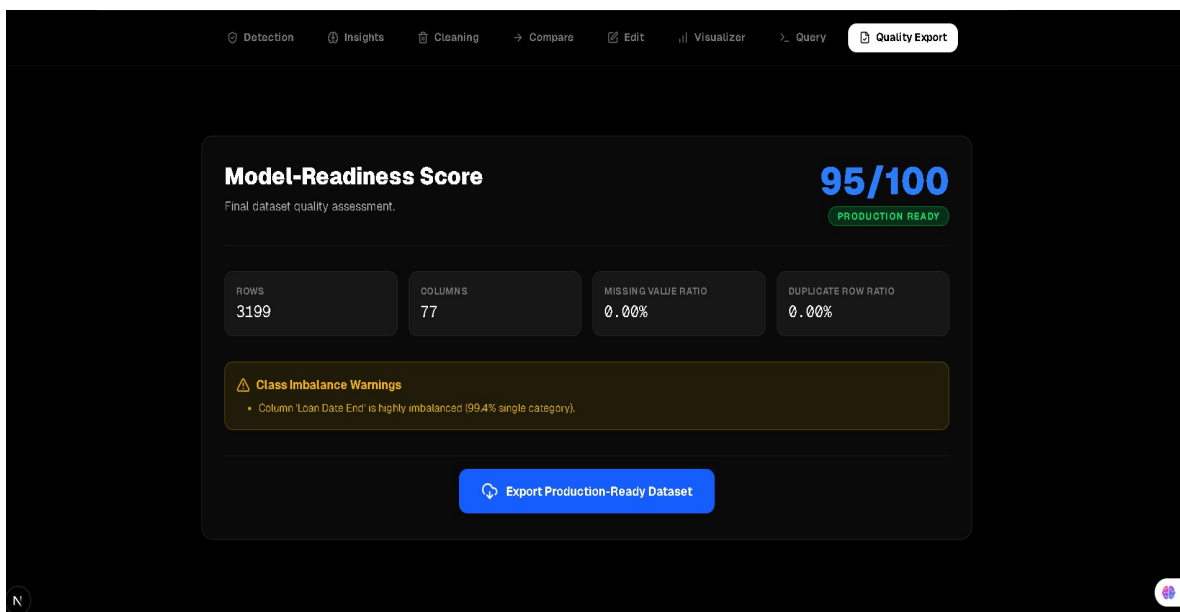


Fig. 5.4.9 Quality Export Page

Above Fig. 5.4.9 shows the quality export page, presenting dataset quality scores along with key evaluation metrics. Users can review final insights and validate results before proceeding further. It enables the generation of a reliable and analysis-ready dataset for export.

Chapter 6

Technical Specification

The development of DataSentinel: Advanced Data Quality Intelligence System is guided by a modern, high-performance technology stack designed to ensure scalability, real-time processing, data security, and an enhanced user experience. The system integrates artificial intelligence, neuro-symbolic reasoning, and high-speed inference infrastructure to deliver robust anomaly detection, automated data cleaning, and intelligent analytics.

Unlike traditional data processing systems, DataSentinel emphasizes explainability, privacy preservation, and interactive intelligence, making it suitable for enterprise-grade applications across multiple domains.

6.1 Frontend Technologies

The frontend of DataSentinel is designed to provide a visually rich and highly interactive user experience. It leverages modern web technologies along with animation frameworks to create a cinematic and intuitive interface.

The system incorporates GSAP (GreenSock Animation Platform) for smooth, scroll-driven animations, enabling dynamic transitions and engaging data storytelling. Additionally, Three.js is utilized to render advanced 3D visual components such as the *PurificationCore* and *NeuralCore*, which represent internal data processing stages in an interactive manner.

To enhance user interaction, Lenis smooth scrolling is implemented, ensuring seamless navigation across different sections of the application. The frontend supports features such as file uploads, real-time analytics display, and interactive dashboards, allowing users to explore insights effortlessly. The frontend is optimized for performance and usability, ensuring smooth interaction even when handling moderately large datasets. It provides real-time feedback to user actions, improving responsiveness and enhancing the overall user experience. The design follows a clean and minimal layout, allowing users to focus on data analysis tasks without unnecessary complexity.

6.2 Backend Technologies

The backend is developed using Python 3.x, which serves as the central processing engine for data analysis, machine learning, and system orchestration.

The architecture follows a modular monolith design, where different functional components such as anomaly detection, logic validation, and NLP processing are organized into independent modules while maintaining efficient internal communication.

The backend is responsible for:

1. Data ingestion and preprocessing
2. Execution of anomaly detection algorithms
3. Neuro-symbolic rule generation and validation
4. Secure handling of sensitive information
5. Integration with high-performance inference systems

This design ensures flexibility, maintainability, and scalability for future enhancements.

6.3 Machine Learning and Explainable AI

The machine learning layer is built using Scikit-learn, along with Pandas and NumPy for efficient data manipulation and computation.

The core anomaly detection mechanism is powered by Isolation Forest, which identifies outliers by isolating anomalous data points in feature space. To enhance transparency, the system integrates SHAP (SHapley Additive Explanations), providing detailed explanations for every flagged anomaly.

This combination ensures that users not only detect anomalies but also understand the underlying reasons behind them, making the system highly interpretable and trustworthy.

Furthermore, the integration of explainable AI techniques enhances transparency by providing interpretable outputs alongside predictive results. This allows users to validate anomaly detection decisions and build confidence in the system's analytical capabilities. The combination of statistical methods and explainability ensures a balanced approach between performance and user trust.

6.4 Neuro-Symbolic Logic Gate

A key innovation in DataSentinel is the Neuro-Symbolic Logic Gate, which combines artificial intelligence with mathematical rule-based validation.

This module dynamically generates dataset-specific rules using AI models and applies them to validate data integrity. The logic gate acts as a filtering mechanism, ensuring that only valid and consistent data passes through the pipeline. In cases where excessive data is rejected, a fail-safe mechanism is triggered to bypass the logic gate, preventing system blockage and maintaining operational continuity. This approach enhances reliability by combining:

1. Data-driven learning (neural networks)
2. Rule-based reasoning (symbolic logic)

By integrating these two paradigms, the system is able to detect complex anomalies while also enforcing strict validation rules.

6.5 High-Performance AI Inference

To achieve real-time processing, DataSentinel integrates Groq LPU (Language Processing Unit) acceleration, running LLaMA 3.3 70B models. Groq LPUs are optimized for ultra-fast inference, enabling millisecond-level response times for:

1. Rule generation
2. Natural language queries
3. Data validation logic

This significantly improves system responsiveness compared to traditional GPU-based inference systems.

6.6 Privacy and Security (PII Protection)

DataSentinel incorporates a local-edge PII (Personally Identifiable Information) scanner to ensure data privacy before any cloud interaction. Sensitive information is:

1. Detected locally using pattern recognition techniques
2. Encrypted using SHA-256 pseudonymization
3. Replaced with secure tokens before processing

This ensures compliance with data protection standards and prevents exposure of sensitive user information.

6.7 Smart Data Cleaning

The system includes an intelligent data cleaning module that automatically handles anomalies based on severity and user preferences.

Key functionalities include:

1. Dropping invalid data points
2. Quarantining suspicious records
3. Providing before-and-after dataset comparisons

This enables users to maintain high-quality datasets without manual intervention.

6.8 Natural Language Query System

DataSentinel integrates a Natural Language Query (NLQ) system, allowing users to interact with data using plain English.

The system interprets user queries and dynamically generates and executes SQL statements, enabling:

1. Data exploration without coding
2. Instant insights and summaries
3. Context-aware responses

This feature significantly enhances accessibility for non-technical users.

6.9 Visualization and Analytics

For data interpretation, the system provides advanced visual analytics, including:

1. Correlation heatmaps
2. Distribution histograms
3. Categorical breakdown charts

These visualizations help users understand data patterns, detect trends, and validate analysis results effectively.

6.10 Reporting and Data Quality Metrics

These metrics provide a comprehensive understanding of dataset quality and help users evaluate the effectiveness of cleaning and preprocessing operations. They also assist in

identifying areas that require further improvement, ensuring better decision-making and reliable analytical outcomes. These reports include:

1. Data completeness
2. Anomaly percentages
3. Integrity scores
4. Cleaning impact analysis

The reports are designed for both technical and business stakeholders, making insights easy to communicate and act upon.

6.11 Development and Deployment Tools

To ensure efficient development, deployment, and system scalability, the following tools are used. These tools support collaboration, performance optimization, and reliable data management across different components of the system.

- 1. Git:** Version control and collaboration
- 2. Redis:** Caching for improved performance
- 3. PostgreSQL:** Structured data storage
- 4. MinIO:** Object storage for datasets

These tools collectively ensure scalability, reliability, and ease of maintenance, enabling the system to handle growing data requirements and support future enhancements effectively.

Chapter 7

Project Scheduling

The project's 14-week phased plan ensured smooth progress from research to deployment by setting clear goals for efficient development, integration, and testing.

Sr. No.	Group Members	Duration	Task Performed
1	Nishigandha, Subodh, Chinmay, Shantaram	1st–2nd Week of January	Defined the project scope, objectives, and system architecture for DataSentinel: Data Quality Intelligence System. Conducted an in-depth literature survey on data quality assessment, anomaly detection techniques, and existing no-code data analysis tools. Identified suitable datasets and finalized technology stack.
2	Subodh, Shantaram	3rd–4th Week of January	Collected and preprocessed datasets for testing and validation. Performed initial data cleaning, exploratory data analysis (EDA), and feature understanding. Defined data quality parameters such as missing values, duplicates, and inconsistencies.

3	Nishigandha	1st Week of February	Designed and implemented AI chatbot functionality using Mistral-7B via Hugging Face. Developed natural language query handling for dataset interaction and generated context-aware responses for user queries.
4	Subodh	2nd Week of February	Developed core backend logic for dataset ingestion. Implemented statistical analysis functions and integrated anomaly detection algorithms (Isolation Forest, LOF, DBSCAN, Z-score).
5	Chinmay	3rd Week of February	Designed and developed Streamlit-based frontend including dataset upload interface, analysis dashboard, visualization panel, and user interaction components. Ensured responsive and user-friendly UI design.
6	Shantaram	4th Week of February	Performed system-level integration, debugging, and optimization for performance and scalability.
7	Nishigandha, Subodh	1st Week of March	Conducted system testing, validation of anomaly detection results, and refinement of data quality scoring mechanisms. Improved explainability of outputs using XAI techniques.

Sr. No.	Group Members	Duration	Task Performed
8	Nishigandha, Chinmay	2nd–3rd Week of March	Prepared project documentation including report writing, PPT development, research paper drafting, and final presentation materials. Ensured proper formatting and academic alignment with university requirements.
9	Nishigandha, Subodh, Chinmay, Shantaram	4th Week of March	Final system testing, debugging, performance optimization, and project submission. Reviewed all modules and ensured system stability and correctness.

Table 7.1 Project Scheduling Details

Above table 7.1 presents the project scheduling and task distribution for DataSentinel, outlining the responsibilities of team members along with the duration and activities performed during each phase of development.

A Gantt chart's visual interface allows you to see Smartsheet™'s details about each task as well as project dependencies.

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34

Chapter 8

Results

This chapter highlights the key outcomes from the design and implementation of DataSentinel: Data Quality Intelligence System, focusing on system performance, analytical results, and user interaction. The platform integrates multiple core modules—including Dataset Profiling, Data Quality Assessment, Anomaly Detection, and Natural Language Query Processing—which were evaluated for accuracy, efficiency, and usability. The dataset profiling module demonstrated fast and reliable performance, generating statistical summaries, identifying missing values, and detecting duplicates within a few seconds of dataset upload. This ensured real-time exploratory data analysis (EDA) capabilities for users without requiring manual intervention. The data quality assessment module successfully computed quality scores based on completeness, consistency, and uniqueness, providing users with a clear understanding of dataset reliability.

The anomaly detection module, which utilizes an ensemble of algorithms such as Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and Z-score, exhibited strong performance in identifying both global and local anomalies across datasets. The system effectively detected outliers in numerical features, inconsistencies in categorical data, and unusual patterns in structured datasets. The ensemble approach improved robustness by minimizing false positives and false negatives compared to individual models. While minor limitations were observed in detecting highly complex or domain-specific anomalies, the overall performance remained consistent and suitable for practical data analysis scenarios. Additionally, the system maintained efficient processing times, with anomaly detection and visualization generation completing within a few seconds for small to medium-sized datasets.

8.1 Comparison With Existing Solutions:

Existing data analysis tools such as spreadsheet software (e.g., Excel), Python-based libraries, and standalone data cleaning frameworks address only specific aspects of the data

analysis pipeline. Spreadsheet tools provide basic data cleaning and visualization capabilities but lack advanced anomaly detection and scalability. Python-based tools such as Pandas and Scikit-learn offer powerful analytical capabilities; however, they require programming expertise, making them inaccessible to non-technical users.

Key Differentiator:

DataSentinel uniquely combines automated data quality assessment, ensemble-based anomaly detection, explainable AI, and natural language interaction into a single integrated platform. This holistic approach not only improves analytical efficiency but also enhances accessibility, making advanced data intelligence available to users without programming knowledge.

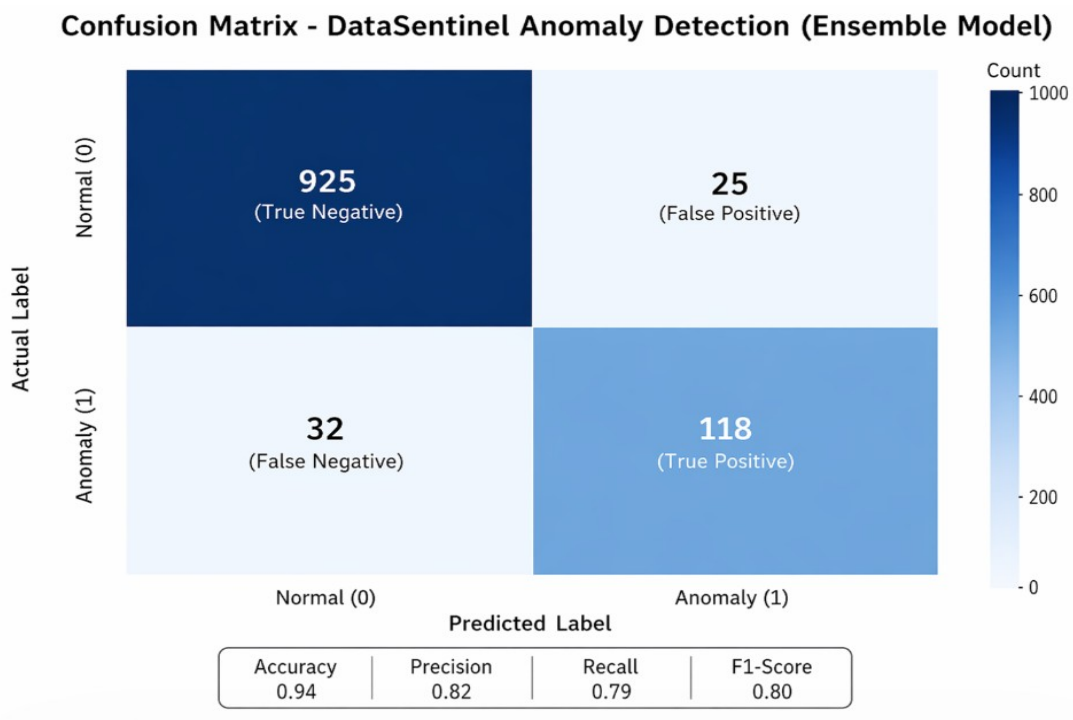


Fig 8.1.1 Confusion Matrix for DataSentinel Anomaly Detection Model

Above Fig. 8.1.1 shows the confusion matrix for the DataSentinel anomaly detection model, illustrating classification performance across normal and anomalous data. It presents counts of true positives, true negatives, false positives, and false negatives. The matrix highlights the model’s effectiveness in accurately detecting anomalies while maintaining low error rates.

Chapter 9

Conclusion

DataSentinel: Data Quality Intelligence System successfully addresses a critical gap in modern data-driven environments by providing an intelligent, unified, and user-friendly platform for data analysis and preprocessing. By integrating automated data profiling, data quality assessment, ensemble-based anomaly detection, and natural language interaction into a single system, it significantly reduces the complexity traditionally associated with data science workflows. The platform empowers users—especially students, analysts, and non-technical individuals—to understand and refine their datasets without requiring programming expertise, thereby democratizing access to advanced data intelligence tools.

The system’s technical foundation, which incorporates machine learning algorithms such as Isolation Forest, Local Outlier Factor (LOF), DBSCAN, and statistical Z-score methods, demonstrates strong capability in identifying anomalies and inconsistencies within datasets. The use of an ensemble approach enhances robustness and reliability by combining the strengths of multiple algorithms. From an architectural perspective, the system leverages a modular monolith design with a Streamlit-based frontend, Python-based backend, and scalable storage solutions such as PostgreSQL and MinIO. The platform also demonstrates strong performance in real-time data analysis, generating insights, visualizations, and reports within a short time frame.

Beyond its technical achievements, DataSentinel aligns with global initiatives such as Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) and SDG 4 (Quality Education) by promoting accessible technology and enhancing data literacy. It contributes to bridging the gap between raw data and actionable insights, fostering informed decision-making and enabling users to build confidence in handling data.

Overall, DataSentinel establishes a strong foundation for future-ready exploratory data analysis (EDA) systems by combining automation, explainability, and usability. While the current system meets its objectives effectively, continuous improvements and integration of advanced capabilities will further enhance its applicability in real-world scenarios.

Chapter 10

Future Scope

This chapter outlines the strategic vision for the future development of DataSentinel, building upon the current platform's capabilities while addressing existing limitations. The system, in its present form, provides a comprehensive solution for data profiling, anomaly detection, and explainable insights; however, there are several opportunities for enhancement to further improve scalability, intelligence, and usability.

One of the key areas for future improvement is the integration of advanced machine learning and deep learning models for more sophisticated data analysis. Incorporating techniques such as AutoML and transformer-based models can enhance pattern recognition, feature extraction, and predictive capabilities. Additionally, extending the anomaly detection framework to support time-series data and streaming data will enable the system to handle real-time analytics, making it suitable for applications such as financial monitoring, IoT systems, and fraud detection.

Another important enhancement involves the integration of real-time APIs and external data sources. By enabling connectivity with databases, cloud platforms, and live data streams, DataSentinel can evolve from a static analysis tool into a dynamic data intelligence platform capable of continuous monitoring and automated decision-making. Furthermore, expanding the natural language processing capabilities to support more advanced conversational analytics and multilingual interaction will improve accessibility and user engagement.

The development of dedicated mobile and cloud-based deployments represents another significant direction for future growth. A mobile application would allow users to analyze datasets and receive insights on the go, while cloud deployment would enable scalable processing of large datasets without hardware constraints. Enhancing the user interface with more customizable dashboards and advanced visualization techniques will further improve user experience and analytical clarity.

In addition, the system can be extended to include collaborative features, allowing multiple users to work on datasets simultaneously and share insights. Integration with version control systems for datasets and analysis pipelines can also improve reproducibility and data governance. Another promising direction is the incorporation of federated learning techniques, enabling data analysis across distributed systems without compromising data privacy.

Finally, continuous improvement of Explainable AI (XAI) capabilities will remain a key focus area. Providing deeper insights into model behavior, confidence levels, and decision boundaries will further enhance transparency and trust. By addressing these areas, DataSentinel has the potential to evolve into a comprehensive, scalable, and intelligent data analysis ecosystem that supports both academic and industrial applications.

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